

TARCU, Romania

WMO: 153170

Lat: 45.2810N

Lon: 22.5328E

Elev: 2182

StdP: 77.72

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.6	-30.7	0.3	-6.9	-26.4	0.4	-9.1	34.0	-10.1	28.5	-11.2	12.4	20	1.037

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.2	16.7	12.3	15.2	11.3	14.0	10.7	13.4	15.5	12.4	14.3	11.5	13.3	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	11.9	14.2	11.6	11.1	13.3	10.7	10.5	12.5	45.1	15.8	42.1	14.3	39.6	13.4	17.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.2	24.2	20.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-8.0	-6.6	-5.8	-1.9	3.4	7.6	8.9	9.8	5.1	1.9	-1.9	-6.0		
	DBStd	7.49	4.70	4.23	4.20	5.06	3.49	3.48	3.34	3.29	4.35	4.91	4.87	4.59		
	HDD10.0	3525	558	466	489	356	205	87	62	43	153	252	358	496		
	HDD18.3	6478	816	699	748	606	463	322	293	264	396	508	608	754		
	CDD10.0	89	0	0	0	0	1	15	27	37	7	2	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	8.3	9.0	9.9	9.0	8.6	7.6	6.7	7.3	6.0	7.8	7.6	9.2	10.7		
(15) Precipitation	PrecAvg	853	51	47	48	63	86	104	111	75	90	66	53	55		
	PrecMax	1276	110	167	115	106	164	225	250	207	193	137	149	119		
	PrecMin	590	17	7	10	11	34	26	38	5	6	3	0	6		
	PrecStd	202	23	34	25	24	35	47	51	48	49	35	36	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	3.4	4.8	9.5	12.7	16.6	18.3	18.6	16.9	13.6	10.1	7.2		
		MCWB	-3.8	-0.2	1.8	5.9	8.9	12.5	13.1	12.8	13.5	5.4	2.7	-1.0		
	2%	DB	0.8	1.4	2.5	7.7	10.8	14.8	16.3	16.9	14.1	11.8	7.6	3.4		
		MCWB	-3.5	-1.5	0.3	3.9	8.1	11.3	12.2	12.1	10.6	5.2	2.1	-1.7		
	5%	DB	-0.8	-0.1	1.0	6.0	9.3	13.7	14.8	15.4	12.0	10.2	5.6	1.0		
		MCWB	-2.4	-1.8	-0.8	3.2	7.2	10.7	11.3	11.3	9.2	4.8	1.3	-2.4		
	10%	DB	-2.0	-1.3	-0.2	4.7	8.1	12.5	13.4	13.9	10.3	8.4	3.7	-0.4		
		MCWB	-3.5	-2.4	-1.4	2.6	6.3	10.2	10.8	10.9	7.6	4.5	1.5	-2.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	1.2	2.2	6.7	10.0	14.5	15.0	14.5	13.3	8.9	4.8	1.2		
		MCDB	-0.2	2.0	4.1	8.7	11.4	15.6	16.9	16.8	16.5	10.7	6.2	1.3		
	2%	WB	-1.1	-0.1	0.9	5.0	8.6	12.7	13.1	13.2	11.0	7.5	3.6	-0.3		
		MCDB	-0.8	1.1	2.1	6.5	10.1	13.9	15.2	15.6	13.4	9.6	4.9	1.1		
	5%	WB	-2.2	-1.1	-0.1	3.9	7.5	11.6	12.2	12.2	9.5	6.3	3.0	-1.3		
		MCDB	-1.7	-0.6	0.8	5.5	8.8	12.9	13.8	14.4	11.5	8.5	4.7	0.5		
	10%	WB	-3.2	-2.1	-1.1	2.9	6.5	10.5	11.3	11.4	8.4	5.3	2.1	-2.0		
		MCDB	-2.4	-1.7	-0.4	4.7	7.7	11.8	12.8	13.4	9.8	7.5	3.5	-0.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	4.8	4.6	4.6	4.0	4.5	4.5	4.2	4.1	4.6	4.5	4.7		
		MCDBR	5.5	5.0	4.6	5.4	5.6	5.9	6.1	5.2	5.6	5.4	6.1	5.3		
	5% WB	MCWBR	4.1	3.9	3.8	4.4	5.2	5.9	5.4	5.2	4.7	5.1	4.3	3.1		
		MCDWR	4.6	4.6	4.8	4.9	5.0	5.5	5.1	4.8	4.7	5.1	4.5	4.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.247	0.268	0.299	0.340	0.332	0.326	0.336	0.346	0.320	0.293	0.272	0.249		
		taud	2.439	2.396	2.413	2.327	2.381	2.415	2.376	2.346	2.417	2.509	2.527	2.486		
	Ebn at Noon	Ebn at Noon	890	927	939	920	935	939	924	902	902	887	853	853		
		Edh at Noon	68	87	100	119	117	113	116	116	98	78	63	59		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.33	2.12	3.25	4.38	5.24	5.90	5.83	5.27	3.75	2.58	1.52	1.10		
		RadStd	0.17	0.27	0.45	0.49	0.48	0.56	0.47	0.52	0.38	0.31	0.20	0.15		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	+0.66	N/A	N/A	N/A	N/A	N/A	N/A	-115	-175	+118	+60

Nomenclature: See separate page